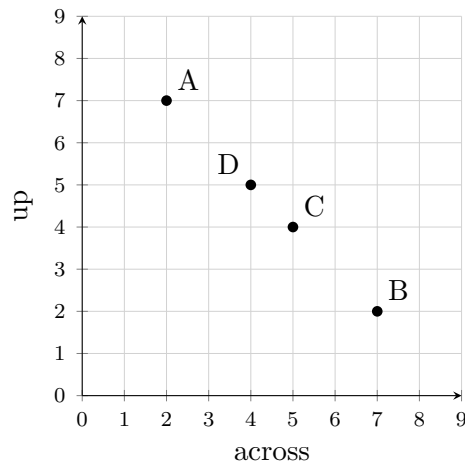


Coordinates, Line Plots and Patterns — Find the Mistake, Then Fix It

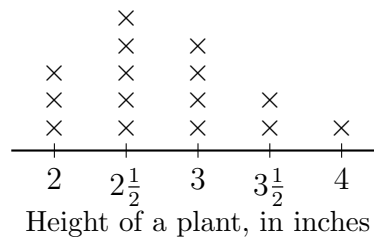
Every mistake on this sheet comes from reading a number in the wrong place. Three rules stop all of them.

- **Across first, then up.** In (x, y) the first number counts across from the corner and the second counts up. $(3, 8)$ and $(8, 3)$ are two different places.
- **On a line plot, count the marks.** The number written under the line says *what* was measured. The marks say *how many*.
- **In a pattern pair, keep the patterns in order.** The first pattern's term always goes in the first slot.

Grid (problems 1 and 2). Four points are plotted and labelled.



Line plot (problems 4 to 6): the height of each bean plant in the class, in inches. One mark $\times$ stands for one plant.



1. Warm-up. Grid. Point C is at (5, 4) and point D is at (4, 5). They are built from the same two numbers but they are not the same place. Moving only along grid lines, how many steps take you from C to D?

Answer: _____

2. Find and fix. Grid. Nia wants to plot (2, 7). She counts 7 across and 2 up instead, and lands on a point that is already labelled.

Which labelled point did Nia land on, and which one did she mean? Then, moving only along grid lines, how many steps apart are the two points?

Answer: _____

3. Find and fix. Patterns. Pattern X starts at 0 and adds 3 each time. Pattern Y starts at 0 and adds 1 each time. Matching terms are written as ordered pairs (X, Y).

Ravi is asked how far *up* the pair with an across-value of 12 sits, and he answers 12. Ravi read the pair the wrong way round. What is the correct up-value?

Answer: _____

4. Find and fix. Line plot. Omar says, “3 plants are 3 inches tall — I read the 3 under the marks.”

Omar copied the height instead of counting the marks above it. How many plants are exactly 3 inches tall?

Answer: _____

5. Line plot. Which height happened most often? Careful: the answer is the height written *under* the tallest stack, not how many marks are in it. Give your answer in inches.

Answer: _____

6. Find and fix. Line plot. Kaya says, “5 plants were measured, because there are 5 different heights marked on the line.”

Kaya counted heights instead of plants. How many plants were measured altogether?

Answer: _____

7. Find and fix. Patterns. Pattern A starts at 1 and adds 2: 1, 3, 5, 7, 9. Pattern B starts at 3 and adds 6: 3, 9, 15, 21, 27. Matching terms make the ordered pairs (A, B) . Sam writes the 4th ordered pair as $(21, 7)$.

(a) What is the 4th term of Pattern A — the number that belongs first? (b) What is the 4th term of Pattern B — the number that belongs second? (c) Write the 4th ordered pair correctly and say what Sam did.

Answer: _____

8. Show what you know.

- (a) On the grid below, plot and label the points $(6, 2)$ and $(2, 6)$.
(b) Write one or two sentences explaining why they are different places even though they are made from the same two numbers.
(c) Write down a way to remember which number goes first, in words a classmate could use.

